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Using Virtual Pets to Increase Fruit and Vegetable Consumption in Children: A Technology-Assisted Social Cognitive Theory Approach

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Abstract

A virtual pet in the form of a mid-sized dog was developed based on the framework of social cognitive theory and tested as a vehicle for promoting fruit and vegetable (F&V) consumption in children. Three groups of children ($N=68$) between the ages of 7 and 13 years were studied: baseline (no treatment), computer only, and virtual dog. Children in the virtual dog condition interacted with the virtual dog for 3 days, setting F&V consumption goals and receiving evaluation and reinforcement based on whether they met their self-set goals. Children vicariously experienced future health outcomes of F&V consumption by seeing, hearing, and feeling their virtual dog's physical and mental health improve or deteriorate based on their F&V consumption in the physical world. Children in the computer only condition interacted with a computer system that presented equivalent features, but without the virtual dog. Children in the baseline condition did not receive any experimental treatment. Results indicated that children in the virtual dog condition chose to be served significantly more F&V than those in the computer only or baseline conditions did. However, children in the virtual dog condition were unable to consume significantly more F&V than those in the computer only condition, although children in those two conditions consumed more F&V than the baseline condition. Food preferences did not differ significantly across the three conditions before and after the experimental treatments. Theoretical and practical potentials of using a virtual pet to promote F&V consumption systematically in children are discussed.

Introduction

Fruit and vegetables (F&V) are higher in water and fiber content relative to energy-dense foods, contributing to lower energy intake and reduction in long-term risks of obesity.[1,2] Despite extensive efforts, <10% of U.S. children and adolescents consume the recommended amount of F&V per day.[3] The ratio of children that fail to meet the recommended amount of F&V more than doubles between the ages of 4 and 13 years as children begin to spend more time away from parents. Consequently, children in this age group should be considered a priority for interventions that promote F&V consumption.

Extending the authors' recent work using virtual pets to promote physical activity,[4,5] the current study developed a virtual pet guided by the framework of social cognitive theory[6] to promote F&V consumption in children between the ages of 7 and 13 years. A preliminary field test was conducted to demonstrate proof of concept at a summer camp. The randomized controlled trial tested a technology-assisted validation system integrated in the virtual pet system, addressed challenges unique to dietary interventions, and compared the efficacy of the virtual pet intervention against a lower-level intervention and a control condition on F&V consumption.

Social cognitive elements of promoting F&V consumption

The features of the virtual pet used in the current study were constructed based on the individual, social, and environment elements of social cognitive theory,[6] designed to motivate change in consumption behavior. Table 1 summarizes how the tenets of social cognitive theory are operationalized in the virtual pet. The virtual pet yields two distinct affordance-based

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Table 1. Operationalization of Social Cognitive Theory Tenets in the Virtual Dog

Virtual dog feature	*Operationalization*	*SCT construct*	*Expected outcome*
Goal setting	The virtual dog asks the child to set a F&V goal for the next meal (e.g., 0%, 25%, or 50% of the plate). The child repeatedly sets a new F&V goal after the meal.	Individual: self-efficacy (confidence that one can consume F&V regularly)	Self-set goals will increase mastery experience when children meet goals. Heightened self-efficacy will motivate children to set incrementally more challenging goals.
Human–animal bond	The human–animal bond is encouraged by allowing children to personalize the dog (e.g., choose a name, choose a collar). The dog is also interactive, responding in real time to the children's verbal and gestural inputs with verbal and gestural outputs. Finally, the virtual dog tracks and responds to each child's progress by syncing its system to the child's unique activity monitor code.	Social: vicarious experiences (Learning through observation). The virtual dog's physical and mental health is linked to the F&V consumption of the child. The child must regularly consume F&V to keep the virtual dog healthy and fit.	Children learn that regular F&V consumption is important by observing health consequences in their virtual dogs.
Interacting with the dog	After every meal, the child has a chance to interact with the dog to assess the dog's physical and mental health. (a) If the child attains the goal, the child will be able to see, hear, and touch the beneficial health outcomes. The virtual dog's heart will be easier to pump and the artery will gain greater elasticity. The virtual dog will be much more responsive to the child's commands as well. If the child fails to attain the goal, the child will be able to see, hear, and touch the detrimental health outcomes. (b) The physical appearance of the dog becomes leaner when children meet more F&V goals and chubbier when children fail to meet goals.	Environmental: incentives (rewards for meeting goals, consequences for failing to meet goals)	The incentive for successfully meeting the goal is designed so that children will be motivated to work hard to unlock the next level of activities with the virtual dog. Consequences will be evident when his/her virtual dog is slow and not responsive when interacting with the child.

SCT, social cognitive theory; F&V, fruit and vegetables.

advantages over traditional means of communicating health information. First, virtual agents are more effective than traditional media in communicating risk by vividly depicting future consequences of present behaviors.[7,8] This is achieved by digitally rendering an accelerated progression of events in the virtual world that might take a significantly longer time in the physical world. As such, the virtual pet could vividly illustrate the positive benefits and negative consequences in the future as a result of present food choices. For instance, when children continue to meet their F&V goals, the virtual pet immediately loses noticeable weight and responds faster than if the children fail to meet those goals. In the physical world, these outcomes would take much longer to manifest, rendering the causal relationship between health behavior and future consequences abstract and difficult for children to perceive.

Second, children are able to experience health outcomes vicariously through the virtual pet. While Bandura[9] stipulates that the vicarious experience from observing others increases one's confidence in engaging in the same behavior (i.e., ''If she can do it, so can I''), he also notes its limitations by claiming that it is a less dependable source of information about one's capabilities than personal experiences are. Virtual worlds may overcome this shortcoming by providing vicarious experiences amplified by sensory-rich information that closely mimic personal experiences.[10,11] In the current study, children vicariously experienced the future health consequences of F&V consumption by examining the cardiovascular health of the virtual pet. In addition to observing the health of the virtual pet's heart and major artery, the children used an interactive force-feedback device to touch and feel the pet's internal organs vividly.

Thus, the combination of these two affordances—in addition to the social cognitive elements of goal setting, evaluation, and reinforcement—is likely to wield greater influence on F&V consumption than not having the virtual pet elements, or not having an intervention at all. The current study compared the virtual pet in promoting F&V consumption against a computer system that offered equivalent goal setting, evaluation, and reinforcement features, as well as a control condition, with the following hypothesis:

H1: The virtual pet system will be more effective in promoting F&V consumption compared with (H1A) a computer system with equivalent features but without the virtual pet, and (H1B) a baseline condition without exposure to any interventions.

Influence of extrinsic goals on food preference

Promoting food consumption presents some unique challenges. In particular, when children are given extrinsic, rather than intrinsic, goals for F&V consumption, wherein F&V intake is described as a means to achieve a goal (i.e., becoming healthy) rather than something naturally enjoyable, the F&V item is considered less preferable.[12–15] For example, when children were told that crackers would make them healthy, they ate fewer and thought the crackers were less tasty compared with children who were given the same crackers without an extrinsic goal.[14] Similarly, children may initially increase their F&V consumption to meet an extrinsic goal (e.g., rewards), but fail to continue the health behavior if their preference for F&V decreases. Thus, measuring change in F&V preferences may be important in forecasting longer-term success of interventions.

Bandura[6] contends that intrinsic motivation is a function of self-efficacy (i.e., perceived confidence in engaging in a behavior) and the ability to evaluate one's abilities. Further research has demonstrated that engaging in mental imagery is helpful in the process of assessing one's level of perceived confidence and actual abilities.[16] Then, the vivid, vicarious experiences of observing and feeling the effects of F&V consumption on the virtual pet may be more effective in promoting intrinsic motivations than mentally imagining the same. These interactive features that drive intrinsic motivations to consume F&V may offset the negative effects of extrinsic goals (i.e., helping the virtual pet become healthier) on food preference:

RQ1: How will the virtual pet system affect F&V preferences compared with the other conditions that do not present extrinsic goals of F&V consumption?

The original virtual pet designed to promote physical activity was created as a kiosk built with consumer grade electronics.[4] Following the guidance of the virtual pet, children interacting with the virtual pet could set physical activity goals, have the virtual pet evaluate the child's physical activity data from an activity monitor, and receive reinforcement in the form of teaching the virtual pet tricks. Children in the comparison condition interacted with a computer system that included the same goal-setting, evaluating, and reinforcing features, but without the virtual pet. Results indicated that children who interacted with the virtual pet engaged in 1.09 hours more of daily physical activity compared with children who interacted with the computer system lacking the virtual pet.

However, scholars recognize that obesity is a multifaceted issue, and several existing studies have used multicomponent interventions to maximize health behavior change.[17,18] Accordingly, this study extended earlier efforts by developing a virtual pet to promote F&V consumption to complement the earlier virtual pet system developed to promote physical activity.

Methods

Participants and design

A sample of 68 children (32 male) between the ages of 7 and 13 years ($M_{age} = 10$ years, $SD = 1.29$ years) was recruited from a camp in the Southeast. To minimize potential contamination across experimental conditions, children were randomly assigned to three cabins, and each cabin was randomly assigned to one of the three experimental conditions—virtual dog, computer only, and baseline—to test the efficacy of the virtual pet in promoting F&V consumption. Children in each cabin moved as a single unit, and different cabins had different time schedules (i.e., meal times) throughout the day.

Children in the virtual dog condition ($n = 25$) interacted with a virtual pet, designed to look like a mid-sized dog, through a desktop computer equipped with the Novint Falcon, a force-feedback joystick that allowed children to touch and feel virtual objects. Each child was given a radio-frequency identification (RFID) card. When the child approached the

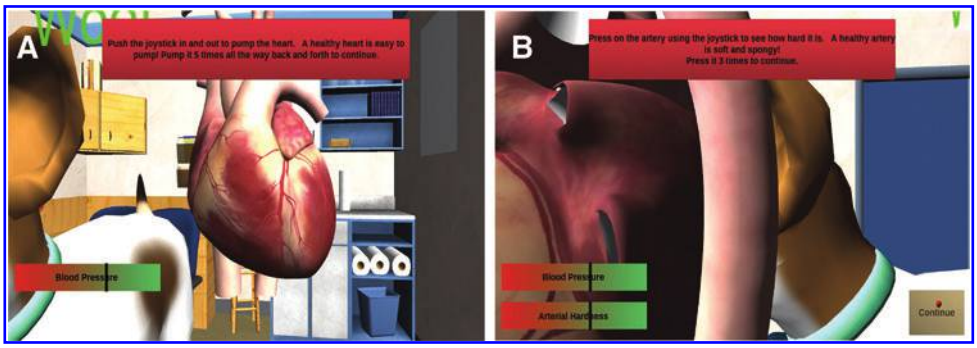

FIG. 1. Children were able to see, hear, and feel the physical health of their virtual dog as a result of their fruit and vegetable consumption in the physical world. The interaction consisted of (**A**) pumping the dog's heart and (**B**) pressing on the dog's major artery to experience vicariously the health consequences of consuming fruit and vegetables. The sliding scale at the lower left-hand corner of the screen indicated the virtual dog's health, ranging from red (unhealthy) to green (healthy) depending on the children's fruit and vegetable goal attainment. Color images available online at www.liebertpub.com/cyber

virtual dog station, the RFID card reader automatically pulled up the child's unique virtual dog. During the first interaction, children personalized their virtual dog (name and collar color) and set their F&V goal—0%, 25%, or 50% of the total food on their plate. The United States Department of Agriculture advises Americans to fill half of their plates with F&V (www.choosemyplate.gov). The virtual dog provided information on the menu items for the subsequent meal and indicated what would (e.g., whole fruit, tossed salad) or would not (e.g., corn, ice cream) contribute toward the F&V goal.

Camp counselors evaluated and recorded F&V consumption using a smartphone application created by the authors, prefilled with demographic information on each participating child, to facilitate the observation and documentation of F&V consumption. The counselors made two observations—one following food serving, and one following actual consumption. This information was immediately reflected on the virtual dog system. When the child approached the virtual dog system after meal time, the system was already updated, and the virtual dog's health changed to reflect each child's F&V goal achievement. Children then (a) received verbal reinforcement (e.g., ''Congratulations! You reached your goal!''), (b) earned points for each goal achieved, and (c) vicariously experienced the effect of F&V consumption on the virtual dog's mental and physical health.

Children assessed the virtual dog's physical health by using the haptic joystick to pump the virtual dog's heart and feel its major artery (Fig. 1). Dogs in poor health as a result of the child failing to meet F&V goals had hearts that were more difficult to pump and arteries with little elasticity (i.e., participants felt resistance through the haptic joystick). Children then assessed the virtual dog's mental health by using the haptic joystick to teach the dog various tricks (Fig. 2). Dogs in better health had faster response times to commands than dogs in poor health did. Children used the points they earned by meeting F&V goals for a chance to teach more sophisticated tricks to their virtual dogs.

In the computer only condition ($n = 22$), children interacted with desktop computers using the same goal setting, evaluating, and reinforcing features. The computer system automatically recognized each child's RFID card and asked the child to set a F&V goal. If the child met the goal, the computer issued the same congratulatory message used for the virtual dog condition; if the child did not meet the goal, the same prompt for negative verbal reinforcement was issued. The only difference from the virtual dog condition was

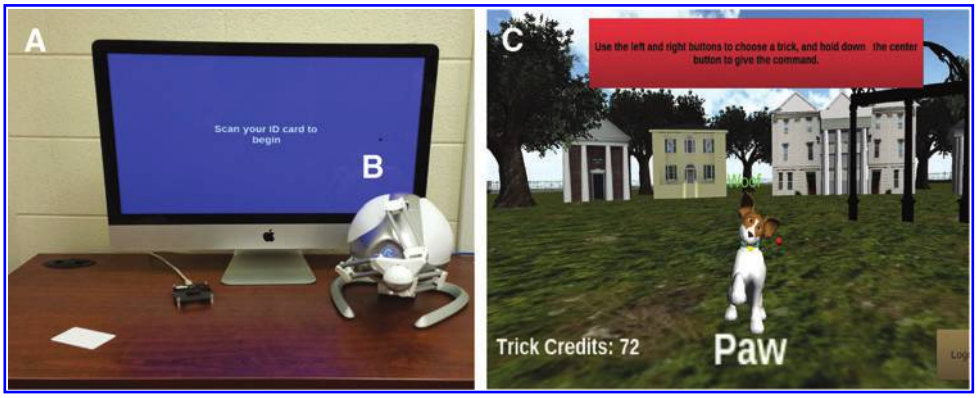

FIG. 2. The virtual pet system employed (**A**) a desktop computer and (**B**) a force-feedback joystick to interact with (**C**) the virtual dog. In addition to the physical health check-up, children were also able to assess the mental health of their virtual dog by using the points they earned from obtaining their fruit and vegetable consumption goals to teach the virtual dog new tricks (**C**). The computer only system shared an identical setup but without the virtual dog. Color images available online at www.liebertpub.com/cyber

that the child did not interact with the virtual dog. Children in the baseline condition ($n=21$) were not exposed to any experimental treatments. As in the other two conditions, their F&V consumption was observed and reported by camp counselors.

Procedure

The experiment began on a Monday afternoon and terminated on the following Thursday afternoon. The first food preference survey and the consent documents were collected prior to the experiment. Each day, children gathered at staggered times based on their experimental conditions to eat breakfast, lunch, and dinner. During meal time, children selected which food item to be served and its amount. The camp counselors were instructed to serve F&V on specific areas of the plate to minimize variance in observations. Counselors made F&V observations immediately after serving and then after consumption for each meal. Children in the virtual dog and computer only conditions went to their respective computer stations for evaluation and reinforcement when they had finished eating. Children in the baseline condition had free time without any experimental treatments. After the third day, all children answered the food preference survey again.

Measures

Served F&V. The amount of F&V served was observed for each meal immediately after the food was served. Similarly, prior studies[14] have assessed the amount of food that children intended to eat as a proxy measure of consumption.

Consumed F&V. The ratio of F&V actually consumed for each meal was observed.

ΔF&V preference. A 9-point interval scale, where 1 = "super bad" and 9 = "super good," measured food preferences for each of the F&V items on the menu. The scale was adopted from earlier scales that measured hedonic food preferences in children.[19] F&V preference was measured once before and once after experimental treatments. The seven items displayed high reliability (Cronbach's $\alpha=0.81$ pre-experiment; Cronbach's $\alpha=0.87$ post-experiment) and were averaged to create an index of F&V preference. The pre-experimental score was subtracted from the post-experimental score to create an index of change in F&V preference.

Results

A univariate analysis of variance (ANOVA) was conducted with the experimental condition as the independent variable and served F&V as the dependent variable. The main effect of the experimental condition was significant, $F(2, 65)=15.68$, $p<0.001$, $\eta^2=0.33$. Post hoc analysis using Tukey's HSD test confirmed that children in the virtual dog condition chose to be served significantly larger portions of F&V ($M=31.51$, $SD=1.23$) than those in the computer only ($M=26.95$, $SD=1.31$, $p=0.04$) or baseline conditions ($M=21.33$, $SD=1.34$, $p<0.01$). The children in the computer only condition also chose to be served significantly larger portions of F&V than those in the baseline condition ($p=0.04$). Based on the amount of F&V the children chose to put on their plates, both H1A and H1B were supported.

A second univariate ANOVA was conducted with the experimental condition as the independent variable and consumed F&V as the dependent variable. The main effect of the experimental condition was significant, $F(2, 65)=19.48$, $p<0.001$, $\eta^2=0.37$. Tukey's HSD test confirmed that children in the virtual dog condition consumed significantly larger portions of F&V ($M=24.05$, $SD=1.40$) than those in the baseline condition ($M=21.80$, $SD=1.53$, $p<0.01$). Children in the computer only condition also consumed significantly larger portions of F&V ($M=22.17$, $SD=1.50$, $p<0.01$) than those in the baseline condition did. However, no significant differences in F&V consumption were identified between the virtual dog and computer only conditions ($p=0.63$). Based on the actual amount of F&V consumed, H1B was supported but H1A was not.

A final univariate ANOVA was conducted to investigate RQ1 with the experimental condition as the independent variable and ΔF&V preference as the dependent variable. The main effect of experimental condition was not significant, $F(2, 65)=1.82$, $p=0.17$, $\eta^2=0.06$. Descriptive statistics for this measure are provided in Table 2. Children in all three conditions had an increase in F&V preferences after the intervention, but the differences between experimental conditions on the change in F&V preferences were not significant.

Discussion

The results of the current study demonstrate the promising potential of virtual pets in promoting F&V consumption in children. When used in conjunction with the earlier virtual pet system for physical activity, the synergistic effect of promoting both physical activity and F&V consumption is likely to have a strong impact on obesity prevention and management in children. This echoes recent suggestions that interactive digital technology may serve an important component of multilevel interventions designed to combat obesity.[8,20–22]

Food consumption measurement based on 24 hour recalls, although widely used, relies heavily on respondents' memory and ability to estimate portion sizes, resulting in inaccurate measures of consumption.[23–25] These challenges would pose even greater problems for children than for the adults that participated in those previous studies. The current study successfully implemented technology-assisted observation measures that were integrated into the virtual pet system. This technology maximizes data quality and consistency without burdening the users. Ostensibly, parents or school personnel may easily take the place of the camp counselors in using the smartphone application to observe and report F&V consumption.

The current study also parsed out the effect of the virtual pet by comparing it with a computer system having the same features but without the virtual pet and a baseline condition. Although children in the virtual dog condition served themselves more F&V than those in other conditions, they

Table 2. Descriptive Statistics for Pre- and Post-Experimental F&V Preferences ($N=68$)

	Baseline	*Computer only*	*Virtual dog*
Pre-preference	4.37 (1.75)	4.95 (1.68)	4.37 (1.65)
Post-preference	6.45 (2.04)	6.54 (1.90)	5.38 (1.82)

Values are presented as mean (standard deviation).

were not able to consume more than the computer only condition. One explanation is that although children wanted to eat more F&V (and thus served the amount of F&V they wished to eat), they may have been too full to finish the food on their plate. The results also suggest that despite sharing the same goal-setting, evaluating, and reinforcing elements of social cognitive theory, interacting with the virtual dog was more impactful on healthy eating choices than using the computer system without the virtual dog.

Changes in food preferences before and after the experimental treatment were nonsignificant across experimental conditions. This implies that the interactions with the virtual pet may have offset negative effects of explicit goals and rewards that focus on F&V as a means of achieving physical and mental health. Open-ended responses from the participants regarding their interactions with the virtual dog seemed to support this proposition, with participants commenting that the virtual dog was ''fun to play with.'' Participants also recognized the association between themselves, their food choices, and the virtual dog's health depicted through vicarious experiences: ''I liked that you could have a dog that is as healthy as you are and that you can train your dog to be healthy.'' They also appreciated the helpful outcomes: ''It helped me eat more fruit and salad than I normally do.''

Although promising, these results remain qualified. First, the children who participated in the current study were appropriate representatives of the target population, but the sample size was relatively small. The fact that significant results with moderate to strong effect sizes were obtained with a small sample indicates strong potentials for the virtual pet to affect children's behaviors positively. Nevertheless, future studies should include participants from a wider range of ages, socioeconomic levels, and family backgrounds to increase the generalizability of the results. Furthermore, although the current study used virtual pets as the vehicle of behavior change, a virtual human tailored to share similarities with each child may also be effective in promoting identification and motivation to model its behavior.[11] Finally, food consumption is notoriously difficult to assess with accuracy and consistency. Although the camp counselors received separate training, individual differences in the evaluation of F&V amounts may have led to some unaccounted variation in the data. Future studies should include a measure of interrater reliability.

In sum, this is one of the first attempts to test the effectiveness of a theoretically driven virtual pet to promote F&V consumption in children. Although the results are preliminary, the study yields meaningful contributions for future research of virtual pets as tools for prevention and intervention in the battle against childhood obesity. When developed in tandem with physical activity promotion, the virtual pet is likely to provide comprehensive and multidimensional support for children who wish to live healthily. With the rapid development of accessible and affordable digital technology, the virtual pet system presents a translatable and scalable solution that requires minimal technical finesse so that children from diverse backgrounds and education levels may benefit.

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Author Disclosure Statement

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